

55th CIRP Conference on Manufacturing Systems Rule-based Decision Support for No-Code Digitalized Processes

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Abstract

Digitalized processes offer various advantages: uniform information flow, traceability of process progress, and enhanced analyzability. While these represent initial benefits from the digitalization of the process, a recurring evaluation and optimization of the process are required to sustain and improve process efficiency. For both non-digitalized and digitalized processes, the challenges for process owners are comparable. However, for digitalized processes, convenient tools facilitate process modifications. Particularly no-code digitalized processes exploit the opportunity for quick and agile process adaption with low efforts but require a structured and systematic method to ensure a quality-driven approach. This paper conceptualizes a rule-based decision support model for process improvement that takes advantage of digital traces recorded during process execution of digitalized processes. On the one hand, process mining techniques support data analysis and assist the extraction of process performance indicators as precursors to predict low process performance. On the other hand, process experts compile a list of recommendations for action in workshops and interviews to counteract low process efficiency. The approach to rule-based decision support developed in the scope of this research merges these two dimensions and constitutes an assistive tool for the continuous improvement of digitalized processes. Initial results of the model in practical application support this assessment and demonstrate further research needs.

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1. Motivation

Companies strive to improve their business processes to achieve long-term competitiveness continuously. A powerful lever is process digitalization, which has seen a significant boost in accelerated deployment, partly owed to the pandemic and its implications for workers' collaboration in remote work. [1] It promises a wide range of benefits: better collaboration through information sharing, increased process transparency and efficiency, up-to-date documentation of the process, higher process reliability, and overall cost reduction. [2]

The digitalization of business processes typically proceeds in two steps: digitization and digitalization. The first refers to the digital mapping of previously analog documents and process steps in a digital workflow. It already boosts the process efficiency as it allows for collaboration and minimizes

media disruption. Process digitalization builds upon this basis and exploits the readily available data to improve process efficiency by data analytics, typically referred to as business intelligence. However, qualified IT staff is required for this digital transition as it depends on the available expertise and IT infrastructure. Typically, an organization's Manufacturing Execution System (MES) serves as the platform for digitalized business processes and requires very detailed change request documentation and high expenses in its deployment. [3] This usually presents organizations with immense challenges, as IT staff is a significant bottleneck, especially for small and medium-sized enterprises. [4] No-code digitalization promises to alleviate some of this dilemma as it enables processes digitization without the requirement of expertise in coding. A no-code platform usually manages process digitalization with a drag and drop interface of preset elements required in process

modeling. Thus, it enables process managers and owners to digitize and iteratively adapt processes while minimizing interface conflicts in communicating specific demands, as seen in conventional approaches. [5]

After digitalization, a process still requires continuous monitoring and iterative improvement to ensure process excellence. Figure 1 presents a process lifecycle model that categorizes this activity within the “process optimization” step. [6] As the usage of workflows generates readily available process data, a databased analytical approach is evident to ensure objectivity and support handling the quantity of data. This paper conceptualizes a rule-based decision support model to assist the process manager in exploiting new analytical options. The overall objective is to produce insights and suggest recommendations to support decision-making during process optimization. To better categorize the model in the scientific discourse, chapter 2 introduces a specific application scenario. Afterward, chapter 3 discusses related work before chapter 4 introduces the approach and decision support model. The concluding chapter 5 summarizes the results and gives an outlook on further research.

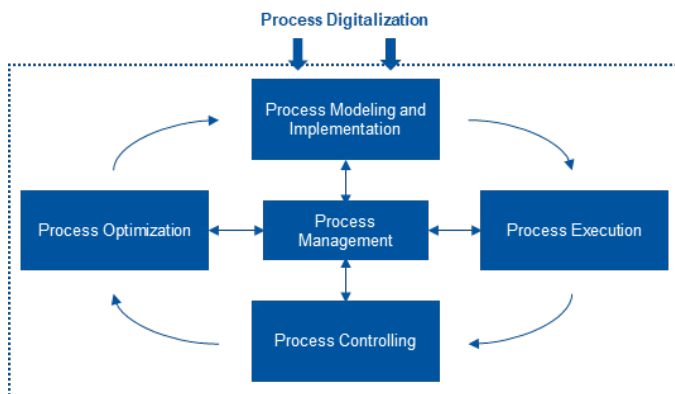


Fig. 1. Process lifecycle model after initial digitalization.

2. Application scenario

Process management focuses on essential processes within an organization for its ongoing operations. Following the standard definition of the New St. Gallen Management Model [7], it differentiates three distinct process types:

- **Management processes:** These cover fundamental normative and strategical processes in an organization.
- **Business processes:** These are value-adding processes with a strong alignment to customer benefits. Generally, all processes linked to value creation fall in this category.
- **Supporting processes:** In-house processes that mainly target internal customers and support (indirectly) in the value creation are considered supporting processes.

Most business processes are integrated into the core IT infrastructure and thus do not require digitalization. The approach presented in this paper mainly addresses supportive processes. While the decision support model’s implications are equally applicable to other process types, their framework

conditions differ significantly. High expenses usually accompany process digitalization. Thus, it is vital to gauge the costs and benefits: management processes involve few people in the organization, thus far worse in terms of benefits than daily-executed processes. Then again, the adaption of core processes demands more resources and a more extended period, as more staff must be involved and updated on any changes. Even minor ones usually require IT expertise and change request documentation, which leaves supporting processes as a primary candidate in the subsequent investigation. Usually, supportive processes are neither in the focus of process management activities nor well documented but are nonetheless essential for the operation of an organization: a well-executed onboarding process accelerates the period until new employees can work autonomously while minimizing misconduct or negligence. Still, few companies produce a well-designed onboarding process. The same applies to the off-boarding of employees to prevent unauthorized access to the organization’s IT systems after their leave.

As few supporting processes have proper documentation, they produce key performance indicators (KPI) to gauge the fulfillment of their objectives and their efficacy and efficiency. Thus, the approach in this paper focuses on process performance indicators, which originate in the process execution and are universally applicable, for instance, metrics about execution time and the number of process loops within one execution instance. These may also serve as prediction parameters to evaluate the status of a process, i.e., whether its efficacy has dropped over the observed period. In addition, the research addresses incremental and continuous process improvement rather than process restructuring because more information not captured within the process data is required to gauge whether a process restructuring is justified.

3. Related work

The review of related work addresses three categories of specification criteria. Their fulfillment assesses the fit within the scope of the previously described application scenario:

- **Object area:** Process optimization addresses the field of incremental process improvement for no-code digitalized processes. It should incorporate the basics of process mining for the databased analytical approach, which uses recorded data in the form of event logs.
- **Target area:** The decision support model draws conclusions based on process performance indicators and connects these with rule-based recommendations for action. The underlying idea of continuous process improvement suggests the consideration of model drift for digitalized processes (process changes recorded in event logs require a process model update in the recurrent application of process mining, similar to concept drift in predictive analytics [8]).
- **General:** Generalizing the described approach to building a decision-support model is mandatory. In addition, a validation with captured data suggests a sound approach.

GROB ET AL. [9] describe an approach for rule-based control of business processes to prevent poor process execution. The underlying idea is the identification of relevant patterns in enriched event logs. The authors develop a rule-based system for process improvement derived from these patterns. While they describe the necessity to consider model drifts and thus changes in the underlying data, it suggests a validation with actual data, but it does not mention it explicitly. However, no specific recommendations follow the established rules.

BOSE ET AL. [10] describe an approach for process improvement for the same processes in distinct organizations, which analyzes process event logs and deploys process mining. It relies on conformance checking and focuses on the process lead-time to quantify process quality. The approach concentrates on statistical correlations between performance indicators and process characteristics to suggest core issues and derives hypotheses. While the validation uses actual process data, it does not extend towards recommendations for action.

YILMAZ AND KARAGOZ [11] present a similar approach that builds upon deviations between similar processes observed in distinct organizations but use a synthetic data set for the validation. The forte of their approach lies in the generalizability of clearly defined and deployed metrics to identify clusters of nonconformance but lack in the description of specific recommendations for action.

EL HADJ AMOR AND GHANNOUCHI [12] discuss quantitative and qualitative key performance indicators and correlate these metrics among one another in conjunction with process steps to derive knowledge about the process execution. A newly introduced ontology to connect this information supports the process manager in the decision-making during process improvement and indicates good generalizability.

CHERNIE ET AL. [13] develop an approach with a high degree of generalizability that uses predetermined target values as a benchmark for performance improvement prior to using process mining for process improvement. Here, a newly added process redesign phase in the process lifecycle considers pattern recognition and process metric calculation methods but does not address them in detail. In contrast, DIBA ET AL. [14] define specific key figures that may serve as an initial starting point for process improvements and recommendations for action. The authors give suggestions on specific activities and considerations during the data analysis and deployment of process mining but refer to a specific application scenario in procurement. CHO ET AL. [15] present a similar approach based almost only on the analysis of event logs but is very specific to applying process mining in the hospital sector. Akin to other authors, they also neglect recommendations for action.

GRAAFMANS ET AL. [16] connect the deployment of process mining with the Six Sigma methodology DMAIC, a common systematic approach for continuous process improvement in quality management. The authors assign process mining techniques to DMAIC phases to expand the toolset with databased analytics. While the described methodology is generally applicable, the shortcoming is the missing disclosure of information on how it can support deriving rule-based recommendations for action.

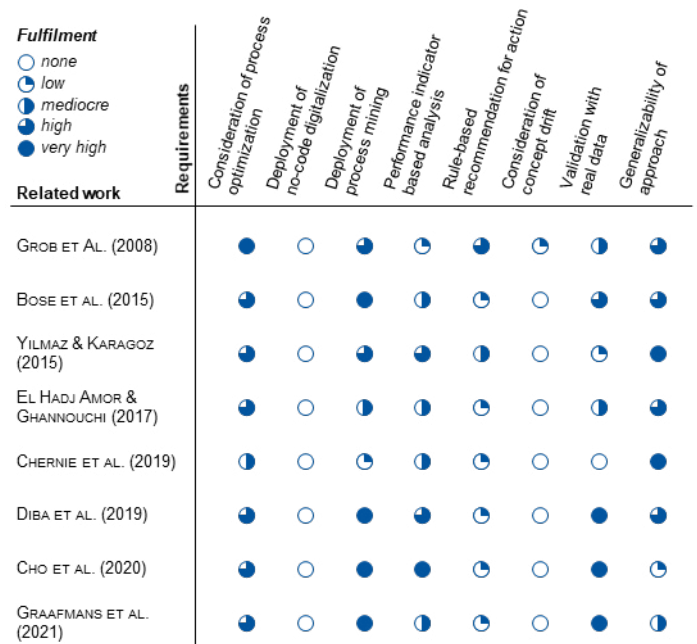


Fig. 2. Overview of related work.

The graphical evaluation in figure 2 summarizes the research on related works about the predefined requirement characteristics and the individual evaluation level. It is evident that most related work neither covers the deployment of no-code digitalization nor considers model drift in their scope of research. Instead, most approaches cover continuous process optimization and deploy process mining as the principal means for analyzing process data and validating their course of action on real data sets. A closer look shows that the combination of an indicator-based analysis and a rule-based decision support scores either mediocre or low on average. This paper conceptualizes a rule-based decision support model based on these gaps in the subsequent chapter.

4. Concept

The concept for the rule-based decision support model intends to give the process manager insights into the process performance of digitalized processes and recommendations for action to support the decision-making in process optimization. Process performance indicators extracted from event logs link to rule-based recommendations for action derived from workshops with process experts to ensure the generalizability of the approach. As it serves as a general framework, it is intended for the model to be applicable for both initial process analysis and subsequent analysis in the monitoring phase. The modeling is oriented towards the de-facto standard CRISP-DM [17] and expands its phases to accommodate for model drift that appears during model usage (cf. chapter 4.7). Extensive literature explains phases 1 to 6 in detail. Thus, the focus lies in delivering additional information relevant to the application scenario (cf. chapter 2) at hand. Figure 3 presents the adopted approach and structures the subsequent subchapters along its phases.

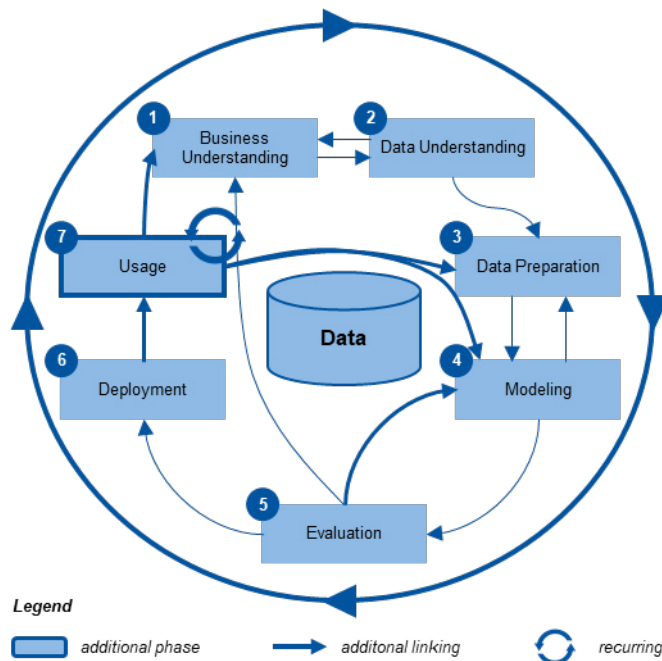


Fig. 3. Modified CRISP-DM.

4.1. Business Understanding

The application scenario described in chapter 2 covers the essential information for business understanding. Figure 4 presents a simplified version of the objective for the modeling. The supportive process of submitting an IT service ticket serves as a consistent example for better comprehensibility. The digital process is deployed in a no-code platform in the software developing entity supporting this research (cf. acknowledgment) and constitutes the evaluation process. It comprises the end-to-end activities of submitting the software product's IT issues to the entities' IT department. Supplementary documentation in the form of a process flow chart or process description supports in clarifying the business understanding but is not necessary here.

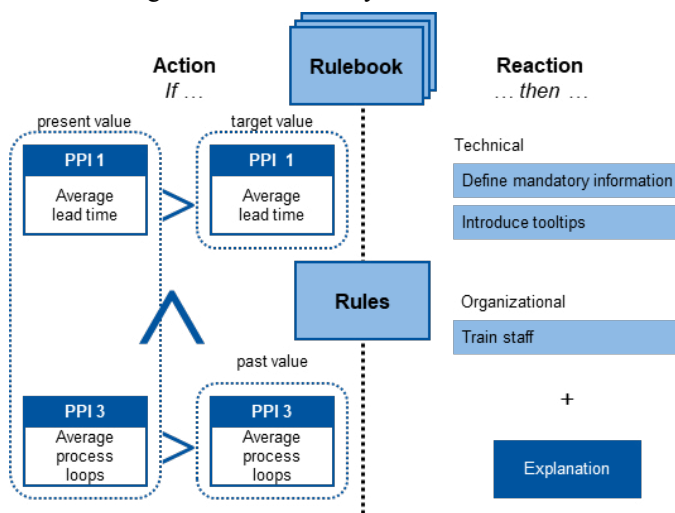


Fig. 4. Relationship between performance indicators, rules and recommendations for action.

4.2. Data Understanding

The premise for the modeling is the availability of process data. As a digitalized process is present and in daily use, its recorded data is available. Table 1 features exemplary data logged in an event log. The information on case ID (a unique identifier for the case), the timestamp for the specified activity, and the employee executing the action constitute the minimal requirements for applying the decision model. Enriching the event log with more data allows for deeper analysis but is not considered in the general approach.

Table 1. Exemplary event log for submitting IT service tickets.

Case ID	Timestamp	Activity	User	Customer	...
0001	1.12.21 8:30	Create ticket	User A	Customer A	...
0002	1.12.21 8:46	Create ticket	User B	Customer A	...
0001	2.12.21 9:14	Accept ticket	User D	Customer A	...
...	...	...	...	...	...

In the case of the consistent example, the event log covers data from 1.12.2020 to 30.06.2021. The log has 22,204 entries assigned to 305 submitted tickets. While the entries appear large in number, only 978 events are relevant to the subsequent analysis after data preprocessing (cf. chapter 4.3) as the no-code platform creates duplicates and non-relevant entries in the log. For example, it logs all interactions with the process, such as opening the application and remaining idle in fixed intervals.

While this covers the basics to derive process performance indicators, creating the list of recommendations for action and the connecting element of rules necessitates more information. Thus expert workshops with process managers, consultants, and quality managers have been conducted between June and October 2021 to compile a universal list of process performance indicators, dependent indicators, and recommendations for action.

4.3. Data Preparation

The initial phase already references core data preparation activities for the event log. However, it extends to further required actions: homogenizing timestamps and their format, allocation of unique process step names, adjusting erroneous representation of values, and anonymization of users. While defining these parameters in the data extraction phase is possible, there was no predecessor case for the employed no-code platform and thus no existing configuration.

The analysis and aggregation of expert workshop results yield 25 process metrics with a brief description, means of calculation, necessary data source, and possible characteristics. Process performance indicators referenced to a specific period allow for many permutations, e.g., the number of average process executions per week or month. The selection of a specific permutation depends on the process; here, most indicators refer to months. The most often stated performance indicators are: lead time, deviation from lead time, number of initiated and closed process instances, number of events, and number of process loops.

4.4. Modeling

While the modeling primarily refers to creating the rule-based decision support model, it also encompasses the deployment of process mining algorithms. As the rule-based system primarily refers to process performance indicators, a visual representation of the process – which is not a mandatory requirement in *Business Understanding*, will support the comprehensibility of the recommendations for action. Process discovery, a process mining technique, complies with the demand for process modeling based on process data and allows for a low-effort model creation. Therefore, the choice for the software environment is ProM as the most distributed open-source framework for process mining algorithms. The process discovery implementation of “inductive visual miner” with default settings creates a sufficient process representation.

The consolidation of the list of performance indicators and recommendations for action constitute the rule-based decision support model (cf. figure 4). Logical operators formulated as rules suggest recommendations for action to the process manager. For example, a good rule is “if average lead time > target lead time, then show recommendation for action (examine process steps with the longest lead time)”. This display incorporates the methods to calculate the desired process performance indicators and dependent indicators. Dependent indicators for the lead time are, for instance, the number of process loops or the number of handover tasks within the execution of one process instance. If a reference value as a benchmark is required, it is possible to predefine these as key performance indicators or reference the average value of past periods. A mockup in Excel serves as the proof of concept. The process manager will not necessarily have insight into the complete list of performance indicators, rules, and recommendations for action but instead, see a curated overview to simplify the analytical results (cf. figure 5).

4.5. Evaluation

Model evaluation with the provided data suggests general feasibility. The metrics calculation appears consistent and realistic, while the proof of concept implemented in Excel yields proper recommendations for action (cf. figure 5). Furthermore, the responsible process manager for the examined process has confirmed the benefits of the visual representation and the insights, as prior to the mockup, no dashboard of any kind was available for the supportive process.

4.6. Deployment

Additional considerations and tasks are required to deploy the proof of concept in the no-code platform. Privacy regulations are essential to anonymize personal information recorded in event logs and access rights management, as process managers and process users require specific information in the analysis. So far, the process manager view is in focus and addresses process monitoring and a backward analysis of processes (diagnostic analysis). In contrast, it is

more interesting for process users to receive suggestions on how to execute a process to increase process efficacy and efficiency (predictive analytics).

4.7. Usage

The adopted CRISP-DM stipulates a phase after the deployment phase to ensure manageability of model drift during model usage: a reference process model created with process mining can lose validity over time due to imminent process changes. This volatility is exceptionally distinct for no-code digitalized processes that feature quick and minor modifications and poor documentation. Over time, these will accumulate and impact the defined rulesets based on deviating initial assumptions, and the model may become less reliable without updates. While the assumption is that concluding the deployment phase yields a feasible and sufficiently performing model for the status quo, continuous process improvement dictates multiple views on the process not necessarily linked to changes in Business Understanding (cf. chapter 4.1). The CRISP-DM cycle already covers model adoption in its cyclical approach for significant changes in frame conditions that result in a relaunch from phase 1. A fine-tuning of, for instance, the rulebook developed in Data Preparation (cf. chapter 4.3) must be possible without traversing all cycles. Another use case is that process indicators may develop in an undesired direction over time. Accordingly, the process manager must check rule triggers. For example, a scheduled change in process performance during company holidays does not require interference. In contrast, a reduction in capacity due to the termination of employees cannot be easily incorporated into the ruleset and constitutes a rare situation.

Another aspect is the implementation of recommendations for action. These directly affect the process performance, but measuring effectiveness is only possible after implementation. Therefore, the time horizon is a significant factor in ensuring that the process is steady. Then again, it is possible that restructuring is required for processes, as incremental changes no longer yield substantial improvements. Figure 6 visualizes the necessary modifications of CRISP-DM Deployment (cf. chapter 4.6) usually does not cover.

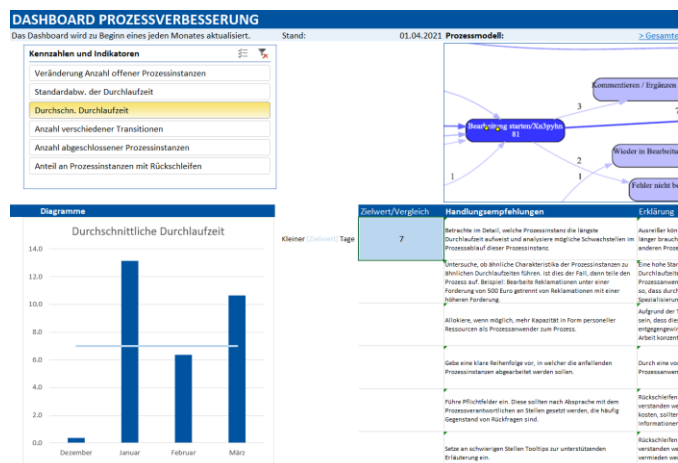


Fig. 5. Proof of concept in Excel tool (in German)

